

Tuning YaPcore (central hub)

One place for day-to-day knobs: `config/` . Optional **fine-tune modules** in `modules/` make each surface discoverable (`provides` / Modules GUI / `FINE_TUNE.txt`) without forking engines — see [MODULES_AND_API.md](#).

Desktop Tune vs in-game admin

Surface	When to use
Swing Tune GUI (desktop)	Edit <code>config/</code> files, Paper templates, module jars — best for cold restarts and packaging
Web dashboard (<code>:8080</code>)	Remote ops: plugins, ranks, regions, map, Discord, console — no SSH
In-game admin menu (<code>/yapadmin</code> , <code>/staff</code>)	Live staff on the server: give items/kits, teleport, moderation shortcuts — see ADMIN_MENU.md

Tune does **not** replace the admin menu; they target different operators and lifecycles.

Layout

Path	Role
<code>config/server.properties</code>	YaP product — ports, dual-stack, Phase 3 flags, packs, JVM
<code>config/paper/</code>	Symlink → <code>folia-kernel/config/</code> — <code>paper-global.yml</code> , <code>paper-world-defaults.yml</code> (high-pop tuned)
<code>config/spigot.yml</code> / <code>bukkit.yml</code>	Symlinks into paper-dir (high-pop tuned)
<code>config/templates/highpop/</code>	Canonical Paper/Spigot/Bukkit performance templates
<code>config/paper-server.properties</code>	Symlink → Paper’s <code>server.properties</code> (<code>view-distance=8</code> , <code>simulation-distance=6</code>)
<code>modules/*.jar</code>	Packaging modules — point at the knobs below (<code>gradle installFineTuneModules</code>)
GUI Tune / Modules tabs	Opens config paths / manages module jars

High-pop Paper configs (product)

YaPcore ships high-pop **starting** defaults on the live kernel. **Entity activation ranges stay uncapped (0)** by default so MSPT vs Leaf/Paper is fair. Tighter EAR lives in

config/templates/highpop-ear/ (opt-in). Packaging module: yap-highpop-module.jar (provides: [highpop]).

Knob	Where	Intent
EAR = 0 (always active)	spigot.yml	Fair vs Leaf — no activation free lunch
Optional tight EAR	templates/highpop-ear/	Production-only, not scoreboard default
Spawn limits / ticks-per	bukkit.yml	Fewer concurrent mobs
ALTERNATE_CURRENT redstone	paper-world-defaults.yml	Cheaper redstone
Hopper / armor-stand / despawn	same	Less TE + entity churn
View 8 / sim 6	paper-server.properties	Less simulated chunks

Farm-heavy worlds that opt into highpop-ear may still need higher villager EAR. Fair MSPT benches also force uncapped EAR themselves.

Created automatically on start (ConfigHub / yap_ensure_config_hub).

```
./scripts/start.sh --fg
gradle installProductDefaults
gradle :gameplay-knobs-plugin:installIntoPlugins # or installGameplayDefaults
```

Paper vs gameplay encyclopedia

Kind	Where
Paper / Spigot / Bukkit	config/paper/..., config/spigot.yml
YaP chassis / network	config/server.properties (+ yap-spatial-module , yap-ops-dashboard-module , ...)
Mob / gameplay encyclopedia	plugins/yap-gameplay-knobs.jar → plugins/YaPGameplayKnobs/knobs.yml (+ yap-gameplay-knobs-module)

What the encyclopedia supports (live)

Feature	How
WASD rideables	ridable: true + controllable: true — passenger Input steers mount every tick

Feature	How
Ridable in water / max Y	<code>ridable-in-water</code> , <code>ridable-max-y</code>
Deep AI	<code>ai.disable-ai</code> , <code>disable-random-stroll</code> , <code>disable-panic</code> , <code>remove-goals: [VANILLA_GOAL_NAME]</code> via Paper MobGoals
Attributes	<code>attributes.max_health</code> , <code>scale</code> on spawn
Breeding delay / water dmg / always-drop-exp / pick-up-loot	Event hooks
Blocks	barrel rows, anvil cost, beehive max bees, crying-obsidian portal frame, lightning-rod range, bed explode

```

mobs:
  cow:
    ridable: true
    controllable: true
    ridable-in-water: false
    ridable-max-y: 320.0
    ai:
      aware: true
      disable-random-stroll: false
      disable-panic: true
      remove-goals: [PANIC]

```

```
/yapknobs reload
```

Original YaP code — not a PurpurMC GPL port. Goal names match Paper `VanillaGoal` constants.